

A Primitive-Ideal Pi-Base Obstruction for C^* -Algebras with BCP

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Source Problem

The source paper is Jinghao Huang, Karimbergen Kudaybergenov, and Rui Liu, *The ball-covering property of von Neumann algebras and noncommutative symmetric spaces*, arXiv:2311.14257. After proving that a commutative unital C^* -algebra $C(K)$ has BCP/UBCP exactly when K has a countable π -basis, the authors ask for a noncommutative version:

$\{U_\alpha\}_{\alpha \in A}$ of K is called a π -basis (weak basis) for K if every nonempty open subset of K contains some U_α in $\mathcal{U}$ [37, Page 15].

In the setting of commutative C^* -algebra, [25, Theorem 2.1] can be reformulated as follows. Note that [25, Theorem 2.1] only deals with real continuous functions. The complex case is a direct consequence of the real case.

Corollary 3.9. *Let $\mathcal{A}$ be a unital commutative C^* -algebra. Then the followings are equivalent:*

- (1) $\mathcal{A}$ has the BCP;
- (2) the pure state space $\mathfrak{P}(\mathcal{A})$ has a countable π -basis;
- (3) $\mathcal{A}$ has the UBCP.

In particular, there exist commutative C^* -algebras containing no minimal projections but having the BCP. For example, the uniform norm closure of the linear span of Haar functions [24, p.150].

We conclude this section by the following question.

Question 3. *It will be interesting to find a non-commutative version of Corollary 3.9, that is, how to characterize those C^* -algebras $\mathcal{A}$ having BCP/UBCP.*

Statement

A family $\mathcal{U}$ of nonempty open subsets of a topological space X is a π -base if every nonempty open subset of X contains some member of $\mathcal{U}$.

Theorem 1. *Let A be a C^* -algebra. If A has BCP, then $\text{Prim}(A)$ has a countable π -base.*

Corollary 1. *If $\text{Prim}(A)$ has no countable π -base, then A fails BCP.*

Corollary 2. *If A has an uncountable family of pairwise orthogonal nonzero closed two-sided ideals, then A fails BCP.*

Idea of the Proof

For an element $a \in A$, the primitive quotients A/P see the norm of a , and the set of primitive ideals where $\|a + P\|$ is almost $\|a\|$ is open. Given countably many proposed BCP centers a_j , collect all

these near-norm open sets. If they are not a π -base, choose a nonempty open set $V \subseteq \text{Prim}(A)$ containing none of them. The ideal corresponding to V contains a norm-one element x , and x vanishes in every primitive quotient outside V . Since each near-norm open set for each a_j has a point outside V , quotient norms force

$$\|x - a_j\| \geq \|a_j\|$$

for every j . Thus the unit sphere cannot be covered by the proposed countable balls.

Preliminaries

We use two standard facts about primitive ideal spaces. First, closed two-sided ideals $I \triangleleft A$ correspond to open subsets

$$U_I = \{P \in \text{Prim}(A) : I \not\subseteq P\}.$$

The open set U_I is nonempty exactly when $I \neq 0$, and every open subset of $\text{Prim}(A)$ has this form.

Second,

$$\|a\| = \sup_{P \in \text{Prim}(A)} \|a + P\|, \quad a \in A.$$

Lemma 1. *Let $a \in A$ and $t > 0$. Then*

$$U(a, t) = \{P \in \text{Prim}(A) : \|a + P\| > t\}$$

is open in $\text{Prim}(A)$.

Proof. Put

$$b = (a^*a - t^2)_+ \in A_+.$$

For $P \in \text{Prim}(A)$, functional calculus in the quotient gives

$$b + P = ((a + P)^*(a + P) - t^2)_+.$$

Hence $b \notin P$ if and only if $\|a + P\| > t$. Therefore

$$U(a, t) = \{P \in \text{Prim}(A) : b \notin P\} = U_{\overline{AbA}},$$

which is open. □

Proof

Proof of Theorem 1. We prove the contrapositive. Assume that $\text{Prim}(A)$ has no countable π -base. Let $(a_j)_{j \geq 1}$ be an arbitrary countable family in A . We will find $x \in S_A$ such that

$$\|x - a_j\| \geq \|a_j\|, \quad j \geq 1.$$

This is enough: any countable family of balls off the origin is contained in the corresponding family $B(a_j, \|a_j\|)$, so no countable ball cover of the unit sphere can exist.

For each j with $r_j = \|a_j\| > 0$, and for every integer m with $1/m < r_j$, define

$$U_{j,m} = \{P \in \text{Prim}(A) : \|a_j + P\| > r_j - 1/m\}.$$

By Lemma 1, each $U_{j,m}$ is open; it is nonempty because primitive quotients norm a_j . The family of all $U_{j,m}$'s is countable. Since $\text{Prim}(A)$ has no countable π -base, there is a nonempty open set $V \subseteq \text{Prim}(A)$ such that no $U_{j,m}$ is contained in V .

Let $I \triangleleft A$ be the nonzero ideal corresponding to V , so

$$V = U_I = \{P \in \text{Prim}(A) : I \not\subseteq P\}.$$

Choose $x \in I$ with $\|x\| = 1$.

Fix j . If $r_j = 0$, there is nothing to prove. If $r_j > 0$, then for every sufficiently large m , $U_{j,m}$ is not contained in V . Choose

$$P_{j,m} \in U_{j,m} \setminus V.$$

Since $P_{j,m} \notin V = U_I$, we have $I \subseteq P_{j,m}$, and hence $x \in P_{j,m}$. Therefore in the primitive quotient $A/P_{j,m}$,

$$x - a_j + P_{j,m} = -(a_j + P_{j,m}).$$

It follows that

$$\|x - a_j\| \geq \|x - a_j + P_{j,m}\| = \|a_j + P_{j,m}\| > r_j - \frac{1}{m}.$$

Letting $m \rightarrow \infty$ gives

$$\|x - a_j\| \geq r_j = \|a_j\|.$$

This holds for every j , so the countable centers (a_j) do not cover S_A by balls $B(a_j, \|a_j\|)$. Since the centers were arbitrary, A fails BCP. $\square$

Proof of Corollary 1. This is the contrapositive proved above. $\square$

Proof of Corollary 2. Let $(I_\alpha)_{\alpha \in \Gamma}$ be an uncountable family of pairwise orthogonal nonzero closed two-sided ideals. The open sets

$$U_\alpha = \{P \in \text{Prim}(A) : I_\alpha \not\subseteq P\}$$

are nonempty. They are pairwise disjoint: if a primitive ideal P belonged to both U_α and U_β , then neither I_α nor I_β would be contained in P ; but $I_\alpha I_\beta = 0 \subseteq P$, contradicting primeness of P .

If $\text{Prim}(A)$ had a countable π -base $\mathcal{B}$, each U_α would contain some $B_\alpha \in \mathcal{B}$. Since the U_α 's are pairwise disjoint, the B_α 's would have to be distinct, impossible for uncountable Γ . Hence $\text{Prim}(A)$ has no countable π -base, and Corollary 1 applies. $\square$

Relation to Earlier Packets

This theorem generalizes the negative half of the continuous-field base obstruction. For commutative $A = C_0(X)$, the primitive ideal space is homeomorphic to X , so Theorem 1 says exactly that BCP forces X to have a countable π -base.

The result is only a necessary condition. It does not prove that every C^* -algebra whose primitive ideal space has a countable π -base has BCP. Simple nonseparable C^* -algebras show why an additional Banach-space or operator-algebraic size condition may still be needed: their primitive ideal space is a point, so the present obstruction is silent.

Verification and Search Bounds

The run indexes were searched for arXiv:2311.14257 together with primitive ideal space, $\text{Prim}(A)$, Glimm space, countable π -base, open ideals, upper semicontinuity, C^* -algebra BCP, and UBCP. Existing packets contained continuous-field and central-base obstructions but no arbitrary-primitive-space necessary condition.

The proof uses only standard primitive-ideal facts: the open-set/ideal correspondence, primitivity implies primeness, primitive quotients norm a C^* -algebra element, and functional calculus in quotients.

Human Review Focus

Check Lemma 1 and the use of the open-set/ideal correspondence in the proof of Theorem 1. Once those standard C^* -facts are accepted, the diagonal escape argument is direct.

References

- [1] Jinghao Huang, Karimbergen Kudaybergenov, and Rui Liu, *The ball-covering property of von Neumann algebras and noncommutative symmetric spaces*, arXiv:2311.14257, 2023.
- [2] Jacques Dixmier, *C^* -algebras*, North-Holland, 1977.
- [3] Minzeng Liu, Rui Liu, Jimeng Lu, and Bentuo Zheng, *Ball covering property from commutative function spaces to non-commutative spaces of operators*, arXiv:2111.04921, 2021.